

Sep 22, 2024 (Due: 08:00 Sep 29, 2025)

1. Design a computational scheme to evaluate $f(x) = \sec x - \cos x$ for a given number $x \in (-1, 1)$. Make sure that your scheme avoids numerical cancellation. You may assume that trigonometric functions are available as library functions.
2. Generate a random vector $x \in \mathbb{R}^2$. Visualize the relative error of

$$x^\top \begin{bmatrix} \cos \theta_k & \sin \theta_k \\ -\sin \theta_k & \cos \theta_k \end{bmatrix} x,$$

where $\theta_k = 2k\pi/2^n$ for $k = 0, 1, \dots, 2^n - 1$. Explain what you have observed.

3. The $n \times n$ Pascal matrix P_n is typically considered ill-conditioned when n is large. Calculate or estimate the condition number of P_n (for an appropriately chosen matrix norm).

4. Assume that $b, \delta b, x, \delta x$ satisfy

$$Ax = b, \quad A(x + \delta x) = b + \delta b,$$

where

$$A = \begin{bmatrix} 987 & 1597 \\ 1597 & 2584 \end{bmatrix}.$$

Construct examples such that

- (1) $\|\delta b\|_\infty / \|b\|_\infty$ is very small while $\|\delta x\|_\infty / \|x\|_\infty$ is very large;
- (2) $\|\delta b\|_\infty / \|b\|_\infty$ is very large while $\|\delta x\|_\infty / \|x\|_\infty$ is very small.

5. Let $\hat{x}$ be an approximate solution of the nonsingular linear system $Ax = b$. The vector $r = b - A\hat{x}$, known as the *residual*, can be treated as a backward error on b because $A\hat{x} = b - r$. Try to construct a matrix E such that $(A + E)\hat{x} = b$ and

$$\frac{\|E\|}{\|A\|} = O\left(\frac{\|r\|}{\|b\|}\right).$$

6. (optional) Show that Wilkinson's rounding model remains valid for complex arithmetic provided that there is no overflow or (gradual) underflow during the calculation. Note that the machine precision u for complex arithmetic is slightly larger than that for real arithmetic.

7. (optional) Provide an rounding error analysis for solving triangular linear system with multiple right-hand-sides.